

Advanced (Habanero)

Q1. What is the absolute value of -5 ?

- (A) -5
- (B) 5
- (C) 0
- (D) 10
- (E) 25

Q2. If $|x| = 3$, what are the possible values of x ?

- (A) $-3, 0$
- (B) $0, 3$
- (C) $-3, 3$
- (D) $3, 6$
- (E) $-6, -3$

Q3. The distance between 0 and -2 on the number line is

$$|0 - (-2)| =$$

- (A) 0
- (B) 2
- (C) -2
- (D) 4
- (E) -4

Q4. The absolute value of x is denoted as $|x| =$

- (A) x if $x \geq 0$
- (B) $-x$ if $x < 0$
- (C) x if $x < 0$
- (D) $-x$ if $x \geq 0$
- (E) $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$

Q5. If $|2x| = 6$, what is $|x|$?

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Q6. The absolute value of $3 - 5$ is

- (A) -2
- (B) 2
- (C) 8
- (D) -8
- (E) 1

Q7. What is $|0|$?

- (A) 1
- (B) 0
- (C) -1
- (D) 10
- (E) -10

Q8. The absolute value $|x - 2|$ represents the distance between x and 2 on the number line. If $|x - 2| = 1$, what are the possible values of x ?

- (A) 1
- (B) 3
- (C) $1, 3$
- (D) $2, 4$
- (E) $0, 4$

Open Ended Questions (FRQ)

Q9. Explain the concept of absolute value using the number line and provide two examples.

Q10. Find the absolute value of x if $|x + 2| = 4$. Provide a numerical answer.

Chef's Tips

- Tip 1: Ensure students understand that absolute value is the distance from zero on the number line.
- Tip 2: Remind students to provide clear explanations and numerical answers where required.
- Tip 3: Encourage students to use the number line to visualize absolute value problems.